

Session 2: Choosing research project

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Overview I

1 Selecting research project

2 Where to find data

3 Break

4 Quarto project

5 Tasks for the week

6 Any questions?

Learning goals of today

- Selecting a good research project

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- Finding data for research project

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- Finding data for research project
- Starting the project in R

Selecting research project

Why a research project

- Intrinsic motivation pivotal

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- Passive learning less effective
- Active learning by doing improves learning outcomes

Learning by doing

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- I want to guide you through the process

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- Empowering you to continue learning

Good research projects

- Aim for a SMART project

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- S pecific

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- **S**pecific
- **M**easurable

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- **R**elevant
- **T**imely

Specific

- Less specific: Has population growth increased?

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- More specific: Has population growth across countries increased the past 10 years?

Measurable

- Less measurable: How has sentiment on inflation changed?

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- More measurable: How have people's responses to inflation questions in standard surveys changed over time?

Achievable

- Less achievable: How have consumption patterns of Germans versus non-Germans changed over the past 40 years using G-SOEP data?

Achievable

- Less achievable: How have consumption patterns of Germans versus non-Germans changed over the past 40 years using G-SOEP data?
- More achievable: What are consumption patterns of Germans versus non-Germans using G-SOEP data?

Relevant

- Less relevant: How often do people in Germany use their car?

Relevant

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- More achievable: How has car use changed in light of the recent increases in fuel costs due to the war on Iran?

Timely

- Less timely: How has the Spanish flue pandemic from 1918-20 changed population at local levels using historical data?

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- More timely: How has the Covid-19 pandemic affected population across the ages using UN population estimates?

Where to find data

Types of data

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 - Administrative data

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- Very expensive if done well

Survey data: Pros and Cons

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CONTRA

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- Can be quite complex
- Often more difficult to gain access

Survey data: Examples

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<https://www.europeansocialsurvey.org/>
- World Value Survey: <https://www.worldvaluessurvey.org/>
- For a range of surveys (in Germany) check out GESIS:
<https://www.gesis.org/angebot/daten-finden-und-abrufen/ueberblick>



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- Be careful of changes in definition of underlying properties (!)

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- May have data gaps or breaks in temporal data

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- OECD:
<https://www.oecd.org/en/data/datasets.html?orderBy=mostRelevant&page=0&facetTags=oecd-languages%3Aen>



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- Example: <https://www.geeksforgeeks.org/data-science/world-bank-dataset-in-r/>

Break

Quarto project

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- Idea is that everything you need to work on your project is in one place
- The project becomes (more) independent from the environment it is executed in
- Enhances portability to other machines (your own or others)

Starting new project

- Create a new project in top-left corner of RStudio

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- I will demonstrate in RStudio

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- Makes sure that research is reproducible

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- If your way of doing things changes the results of your calculations each time you do it that is very bad
- Reproducibility and transparency also (!!!) for your future self

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 - Just make sure to use the correct or /
- Use clearly identifiable folder names, such as \data or \output

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- Output files (such as PDF, HTML, PNG etc.)

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- Notes

Example header

```
#-----  
#  
# Name: Data preparation  
# Purpose: Preparation of OAA2 data analysis  
# Data: OAA2 data, collected the in May and June 2024  
# Author: Heiner Salomon (heiner.salomon@uni-bremen.de)  
# Date created: 12/05/24  
#  
# Notes -----  
#  
# Preparation -----
```



In Quarto: YAML header

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- Possible content:
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 - format: pdf

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- For great overview, see:

<https://charlotte.quarto.pub/cascadia/2-yaml-markdown-basics/2-yaml-markdown-basics.html#/title-slide>



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 - ...
- For great overview, see:
<https://charlotte.quarto.pub/cascadia/2-yaml-markdown-basics/2-yaml-markdown-basics.html#/title-slide>
- For more details, see here: <https://quarto-tdg.org/vaml>



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- Write the progress report by Friday at 15:00

Reading for next week

- Gentzkow, M. and Shapiro, JM. (2014): Code and Data for the Social Sciences: A Practitioner's Guide <https://web.stanford.edu/~gentzkow/research/CodeAndData.pdf>

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- Gentzkow, M. and Shapiro, JM. (2014): Code and Data for the Social Sciences: A Practitioner's Guide <https://web.stanford.edu/~gentzkow/research/CodeAndData.pdf>
- We will discuss those ideas at the beginning of next week's class

Any questions?